Grapefruit Protocol

v1.1.1

COMP 3980 / Set 3O

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Approval

Lead Designers:

We, the lead designers of the Grapefruit Protocol hereby approve this document, and the solidification of version 1.0 of the protocol as defined by this document.

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Calvin Rempel
Lead Designer

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Lead Designer

Project Managers:

We, the project managers of the first generation of Grapefruit Protocol implementations hereby approve this document, and agree to abide by the rule set described therein until such a time that a document describing a new rule set version is approved.

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Description

Grapefruit is a half-duplex, character-based protocol defining a method for communicating ASCII text between two stations. It uses a fixed-sized packet with a single control character determining future intention, a fixed-size data section, and a 32 bit CRC.

The protocol demands that stations request the link before sending data by transmitting an ENQ control character, and that the intended receiver acknowledge the request by sending an ACK character. If the transmitter does not receive an ACK after sending an ENQ within the allotted time (TO2), the transmitter must re-request the line.

Once a station has begun sending data, it is able to send up to a predetermined number of packets (MAX_SENT) defined by the protocol. Once the number of transmissions has been exhausted, or the transmitter has exhausted its data, both stations are again given the opportunity to request the link.

The receiver must validate all packets to ensure the data was not corrupted in transmission. If the data received was not corrupted, the receiver must transmit an ACK.

If corruption did occur, the receiver must reject the packet and transmit a NAK. If the transmitter does not receive the ACK within the allotted time (TO3), or receives a NAK, it must re-send the previous packet. This may be done up to a count (MAX_MISS) specified by the protocol. If MAX_MISS is reached, the transmitter must stop sending and re-request the link after a randomized timeout (T04). The receiver must timeout (TO1) at the same time and may then request the link.

Grapefruit also allows a receiver to immediately reverse the direction of the half-duplex link by sending a single control character. In this event, the transmitter must acknowledge the request and begin receiving data.

Reserved Characters

Name	Hex Value	Decimal Value	ASCII Value
Enquire	05	05	ENQ
Acknowledge	06	06	ACK
Negative Acknowledge	15	21	NAK
End of Text Block	17	23	ETB
End of Text	03	03	ETX
End of Transmission	04	04	EOT
Reverse Interrupt	11	17	DC1
Syn1	12	18	DC2
Syn2	13	19	DC3

Description of Usage

ENQ - transmitted by a station to indicate intent to begin transmitting data.

ACK - transmitted by a receiving station to inform the transmitter that (a) an ENQ request has been accepted, or (b) a data packet has successfully been transferred.

NAK - transmitted by a receiving station to inform the transmitter that the received data packet was corrupted in transmission.

ETB - transmitted by a transmitting station as the control character in a data packet to inform the receiver that more data packets will be transmitted.

ETX - transmitted by a transmitting station as part of the Data Block portion of a data packet to denote the end of meaningful data. ETX is only added if there is less data to transmit than there is room in the packet. ETX is added immediately after the last data character and the remainder of the Data Block may be padded with miscellaneous characters.

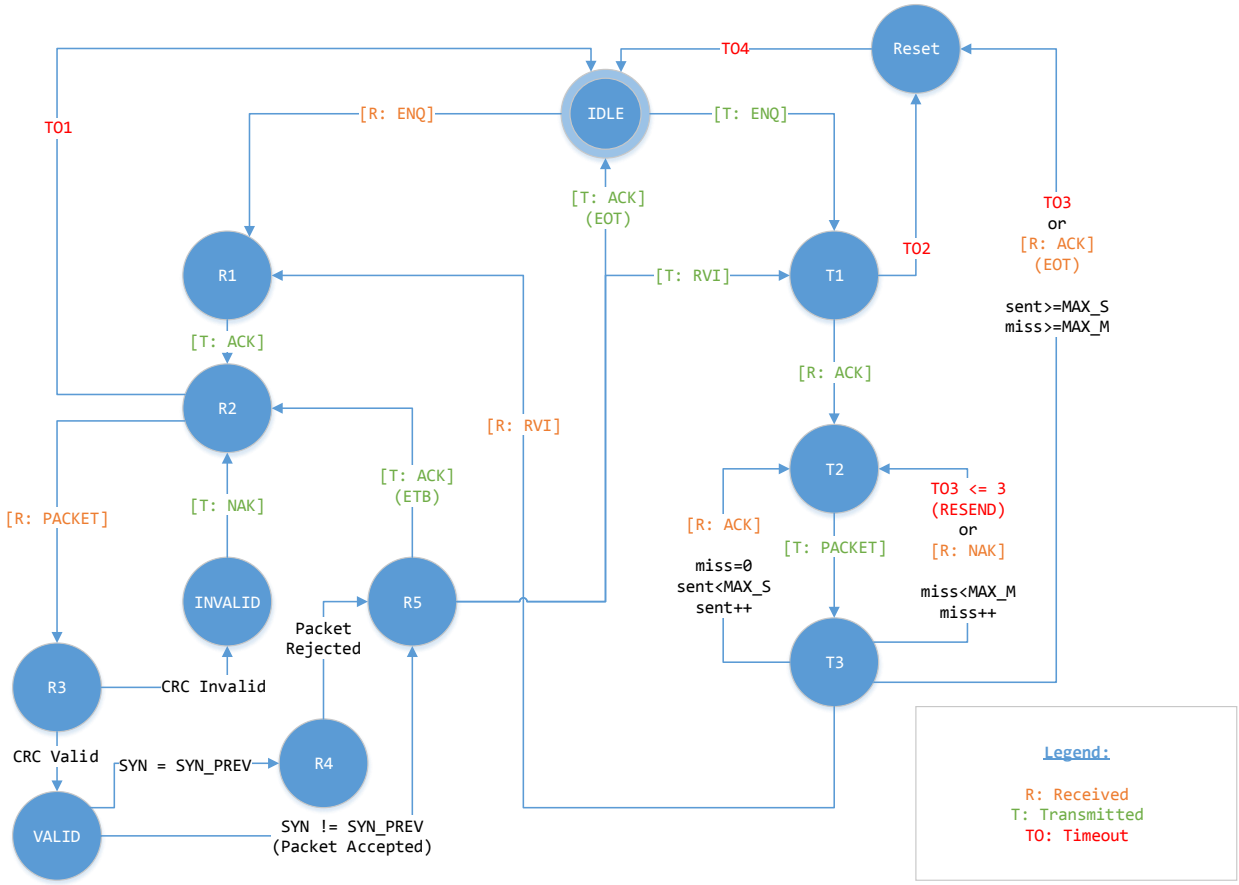
EOT - transmitted by a transmitting station as the control character in a data packet to inform the receiver that no more data packets will be transmitted until the station re-requests the link.

RVI - represented by the ASCII character 'DC1', this character is transmitted by a receiving station to reverse the roles of the stations, allowing the receiver to become the transmitter in the event that priority data needs to be transmitted.

Syn1 – represented by the ASCII character ‘DC2’, this character is sent in every other packet, alternating with Syn2.

Syn2 – represented by the ASCII character ‘DC3’, this character is sent in every other packet, alternating with Syn1.

State Diagram



Legend:

R1 = Received ENQ, Sending ACK State
R2 = Waiting for Packet State
R3 = Received Packet State, Validating
R4 = Duplicate Packet Received
R5 = Reset Receiver

T1 = Sent ENQ, Waiting for ACK State
T2 = Received ACK, Sending Packet State
T3 = Waiting for ACK State

Packet Composition

1024B					
2B		1018B			4B
ETB or EOT	Syn1 or Syn2	DATA	ETX	PADDING	CRC
Header		Data Block			Validation

A Grapefruit compliant packet consists of 3 sections:

1. A header character
2. A data block
3. A validation block

Header:

The 'header' of the packet is two characters. This represents the beginning of the packet. The first character must be ETB or EOT. The character received determines the action to take after receiving and validating the packet. If an ETB is received, then the receiver should expect another packet to be sent after this packet has been acknowledged. The second character must be

either Syn1 or Syn2. The transmitter must alternate between Syn1 and Syn2 for each unique packet. Resent packets must contain the same Syn character as the original. This is to prevent the receiver from accepting the same packet twice in the event that an ACK from the receiver is not received by the transmitter even though the packet was successfully transferred. This block is 2 Bytes long.

Data Block:

The 'data block' contains the payload of the packet, the actual data being communicated. If the data to be sent is less than the size of this block, then an ETX character is to be placed at the end of the data, followed by 'padding characters' (any character) until the length of the block is reached. This section is 1019 bytes long.

Validation Block:

The 'validation block' consists of a 4 byte (32 bit) cyclic redundancy check. This is generated by applying CRC with a polynomial of 0x04C11DB7 to the entire data block (all 1019 bytes). Once the packet is received, the data block is matched against the supplied CRC. If valid, the system acknowledges and

processes the data. If invalid, the system sends a negative acknowledgement and requests a re-transmission. This block is 4 bytes long.

Constants and Formulae

Constants:

MAX_MISS = 3

The maximum number of attempted re-transmissions after transmitted packet was not acknowledged as received.

MAX_SENT = 10

The maximum number of packets that can be sent sequentially before an enquiry must be made to continue.

Timeouts:

TO1 = TO3 * MAX_MISS

The amount of time a receiver should wait before returning to an idle state after acknowledging and enquiry if no packet is received. Must be equal to TO3 times MAX_MISS (i.e. (1200B / byte-rate) * 3)

TO2 = 5B / Current byte-rate

The amount of time a transmitter should wait before returning to an idle state after sending an enquiry if no acknowledgement is received. Equal to the time it takes to transmit 5 data bytes under the current byte-rate. Must be greater than the maximum value of TO4 (i.e. > 4B / byte-rate).

TO3 = 1200B / Current byte-rate

The amount of time that a transmitter should wait before attempting re-transmission of a packet after transmitting a packet. Equal to the time it takes to transmit 1200 data bytes under the current byte-rate.

TO4 ~ U([1B,4B]) / Current byte-rate

The amount of time that a transmitter should wait before returning to an idle state after transitioning to a reset state. Determined dynamically as a random value. Uniformly distributed between the time it takes to send between 1 and 4 data bytes under the current byte-rate.

Suggestions:

Suggested Byte-rate: 16kBps

Suggested Bit-rate: 128kbps

Sample Transmissions

Typical Transmission of < MAX_SENT Packets

ENQ		1.[ETB,Syn1|DATA|CRC]		2.[EOT,Syn2|DATA|CRC]	
	ACK		ACK		ACK

MAX_SENT Reached

ENQ		1.[ETB,Syn1|DATA|CRC]	
	ACK		ACK

x9

	ACK		ACK
ENQ		11.[EOT,Syn1|DATA|CRC]	

MAX_MISS Reached

ENQ		1.[ETB|DATA|CRC]		1.[ETB|DATA|CRC]		1.[ETB|DATA|CRC]		ENQ	
	ACK		NAK		NAK		NAK		...

- All Packets contain “Syn1” in the header.

Receiver Timeout

ENQ		1.[ETB,Syn1|DATA|CRC]		1.[ETB,Syn1|DATA|CRC]		...
	ACK		*TO1*		ACK	

RVI Transmitted

ENQ		1.[ETB,Syn1|DATA|CRC]		ACK		ACK	
	ACK		RVI		2.[ETB,Syn1|DATA|CRC]		...

Revision History

Date	Version Number	Changes
Nov 13, 2014	1.1	• Added Syn1 and Syn2 to Packet Composition • Updated State Diagram to reflect inclusion of Syn1 and Syn2 • Updated Sample Transmissions to include Syn1 and Syn2 • Added Placeholder signature for unrepresented team.
Nov 14, 2014	1.1.1	• Updated State Diagram to show how Syn characters in the packet are used to determine if a packet is duplicate or unique.